PROGRESSIVE WATER TARIFF ANALYSIS

Case Study Report

This Python program calculates water bills for multiple households using progressive usage slabs, a fixed fee, and an additional charge for excessive usage.

1. Problem Statement

Develop a Python program to calculate the water bill for each household based on the number of units consumed. The program must apply different rates to different usage slabs, add a fixed fee, and apply an extra charge when usage exceeds 200 units. It should display the usage and final bill for each household and flag excessive usage.

2. Objective

The objective is to use Python lists, loops, arithmetic operators, and conditional statements to calculate water bills fairly according to progressive tariff rates and identify high water consumption.

3. Assumptions

• The usage values for the four households are 50, 120, 180, and 250 units.

• A fixed fee of Rs. 50 is added to every household's water bill.

• For up to 100 units, the rate is Rs. 2 per unit.

• For 101–200 units, the first 100 units cost Rs. 2 each and each additional unit costs Rs. 3.

• For usage above 200 units, the first 100 units cost Rs. 2 each, the next 100 units cost Rs. 3 each, and each unit above 200 costs Rs. 5.

• Usage above 200 units attracts an additional Rs. 100 excessive-usage charge.

• All amounts are expressed in Indian rupees (Rs.).

4. Algorithm

1. Start the program and initialize the usage list and fixed fee.

2. Repeat the following steps for each household's water usage.

3. If usage is 100 units or less, calculate the charge at Rs. 2 per unit.

4. If usage is between 101 and 200 units, charge Rs. 2 for the first 100 units and Rs. 3 for each additional unit.

5. If usage is above 200 units, charge Rs. 2 for the first 100 units, Rs. 3 for the next 100 units, and Rs. 5 for every unit above 200.

6. Add the fixed fee of Rs. 50 to the calculated bill.

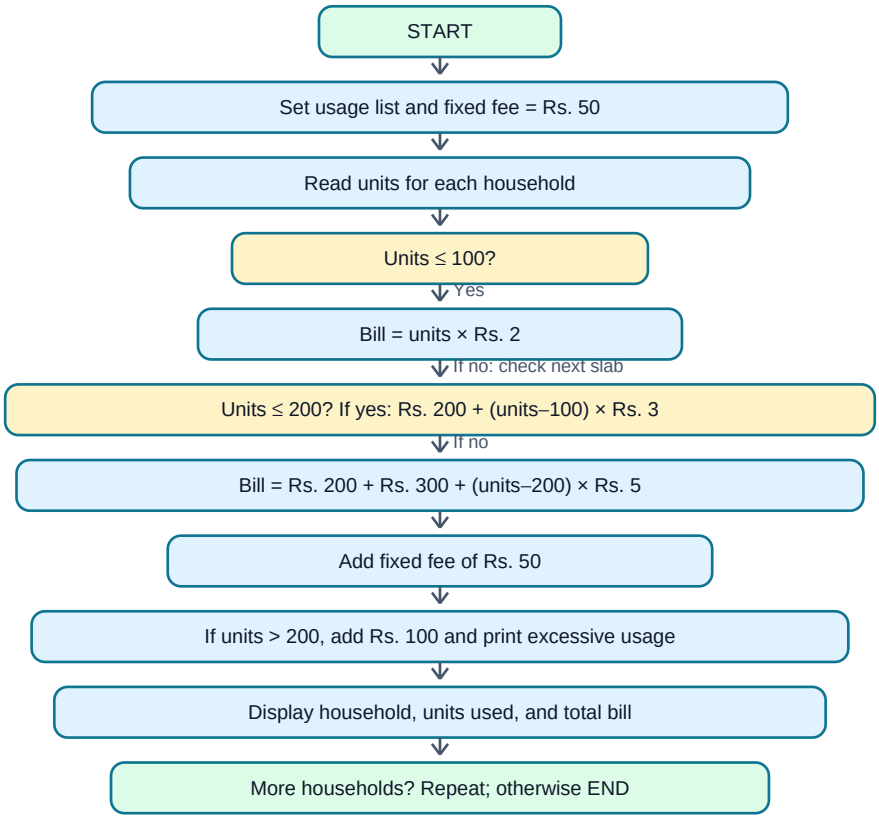
7. If usage exceeds 200 units, add Rs. 100 and print an excessive-usage warning.

8. Display the household number, units used, and total bill.

9. Continue until all households are processed, then stop.

5. Flowchart

The flowchart shows how the program selects a tariff slab, adds applicable fees, and displays the bill for each household.



6. Pseudocode

```
BEGIN
    usage ← [50, 120, 180, 250]
    fixed_fee ← 50

    FOR each household i
        units ← usage[i]

        IF units <= 100 THEN
            bill ← units × 2
        ELSE IF units <= 200 THEN
            bill ← 100 × 2 + (units − 100) × 3
        ELSE
            bill ← 100 × 2 + 100 × 3 + (units − 200) × 5
        END IF

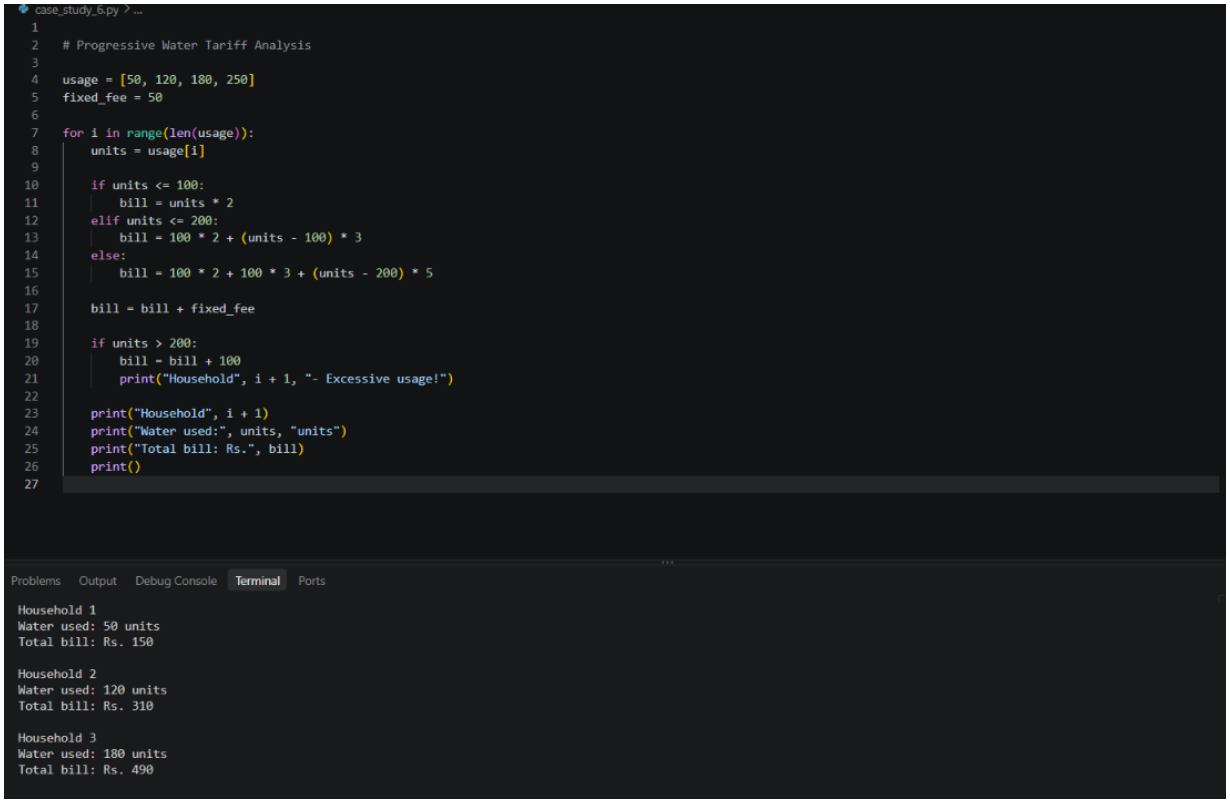
        bill ← bill + fixed_fee

        IF units > 200 THEN
            bill ← bill + 100
            PRINT "Excessive usage"
        END IF

        PRINT household number, units, bill
    END FOR
END
```

7. Screenshot of Python Source Code and Output

The screenshot below contains the supplied Python source code and terminal output. The visible output shows the first three households; the program's usage list also includes a fourth household with 250 units.



8. Conclusion

The Progressive Water Tariff Analysis program calculates bills using progressive rates and applies a fixed fee to each household. It also adds an extra charge and warning when consumption exceeds 200 units. In the screenshot, the visible bills are Rs. 150 for 50 units, Rs. 310 for 120 units, and Rs. 490 for 180 units. For 250 units, the program's stated rules produce a total of Rs. 900. This case study demonstrates how Python can automate water billing and highlight excessive usage.